Deployment Document

Game of Games – Group 1
Liu Restrepo Sanabria, Zoe Kuehne, Thomas Roberts, Jake Acquarulo
April 29, 2026

1 Prerequisites

Before downloading and running the Game of Games, ensure your system meets the following requirements:

- **Java Development Kit (JDK)**: Version 11 or higher
- **Operating System**: Windows, macOS, or Linux
- **Terminal/Command Prompt**: Required for compilation and execution
- **Git (optional)**: For cloning the repository

Verify Java Installation

Open a terminal and run:

```
java -version
```

If Java is not installed, download it from Adoptium or Oracle.

2 Downloading the Code

2.1 Option 1: Clone via Git (Recommended)

```
git clone https://github.com/tzrob06/GameofGames.git
cd GameofGames
```

2.2 Option 2: Download as ZIP

1. Navigate to `https://github.com/tzrob06/GameofGames`
2. Click the green **"<> Code"** button
3. Select **"Download ZIP"**
4. Extract the ZIP file to your desired location
5. Open a terminal and navigate to the extracted folder

3 File Structure

```
GameofGames/
|-- CoinFlip.java
|-- CoinFlipTest.java
|-- EvenOdd.java
|-- EvenOddTest.java
|-- FindTheThimble.java
|-- FindTheThimbleTest.java
|-- FindtheRedThread.java
|-- FindtheRedThreadTest.java
|-- GuessTheNumber.java
|-- GuessTheNumberTest.java
|-- GetInput.java
|-- PlayGames.java
|-- PlayGamesTest.java
|-- README.md
```

4 Compilation

Navigate to the project directory and compile all Java files:

```
cd /path/to/GameofGames
javac *.java
```

Note: If you encounter errors, ensure all files are in the same directory and Java is properly installed.

5 Running the Program

5.1 Standard Mode (Player Mode)

```
java PlayGames
```

When prompted, enter **N** for regular gameplay.

5.2 Test Mode (Debug Mode)

```
java PlayGames
```

When prompted, enter **Y** to enable Test Mode. Test Mode reveals hidden information (e.g., coin flip results, thimble location, red spool position) to assist with testing.

5.3 Running Individual Games (Standalone Testing)

To test a single game without the main menu:

```
java CoinFlip         % Play Coin Flip only
java FindTheThimble   % Play Find the Thimble only
java GuessTheNumber   % Play Guess the Number only
java EvenOdd          % Play Even and Odd only
java FindtheRedThread % Play Find the Red Thread only
```

6 Running Unit Tests

All test classes are standalone Java programs that validate game functionality without requiring JUnit.

6.1 Running a Single Test Class

```
java CoinFlipTest
java EvenOddTest
java FindTheThimbleTest
java FindtheRedThreadTest
java GuessTheNumberTest
java PlayGamesTest
```

6.2 Running All Tests

```
for test in *Test.class; do
    java ${test%.class}
done
```

Or compile and run individually:

```
javac *.java
java CoinFlipTest
java EvenOddTest
java FindTheThimbleTest
java FindtheRedThreadTest
java GuessTheNumberTest
java PlayGamesTest
```

6.3 What the Tests Verify

Each test class validates:

- **CoinFlipTest**: Round range constants, HEADS/TAILS constants, odd number validation, playGame execution
- **EvenOddTest**: Odd round input, even number rejection, out-of-range rejection, invalid guess handling
- **FindTheThimbleTest**: Round range constants, LEFT/RIGHT constants, odd number validation, class instantiation
- **FindtheRedThreadTest**: Valid spool pull inputs, invalid input handling (zero)
- **GuessTheNumberTest**: Valid guess handling, out-of-range rejection, win message display, test mode functionality
- **PlayGamesTest**: Menu navigation (choice 3 enters GuessTheNumber), invalid menu input handling (choice 8)

7 Configuration

7.1 Adjusting Game Settings

Game constants can be modified directly in the source files:

Game	Constant	Default Value
CoinFlip	MIN_ROUNDS, MAX_ROUNDS	1, 9
FindTheThimble	MIN_ROUNDS, MAX_ROUNDS	1, 9
GuessTheNumber	RANGE_GENERATION_MIN, RANGE_GENERATION_MAX	1, 50
EvenOdd	COMPUTER_NUMBER_MAX, PLAYER_NUMBER_MAX	5, 10
FindtheRedThread	TOTAL_SPOOLS	50

7.2 Enabling/Disabling Test Mode

Test mode can be enabled:

- **At runtime**: Respond `Y` when prompted at program start
- **By default**: Set `PlayGames.testMode = true;` in `PlayGames.java`

8 Troubleshooting

Error	Solution
`javac` not recognized	Install JDK and add it to your PATH
ClassNotFoundException	Ensure all `.class` files are in the same directory
Cannot find symbol	Compile all `.java` files together: `javac *.java`
Test class will not run	Ensure the test class has a `public static void main(String[] args)` method

9 Support

For questions or issues:

- GitHub Issues: https://github.com/tzrob06/GameofGames/issues
- Contact the team via Slack (channel: `#group1`)